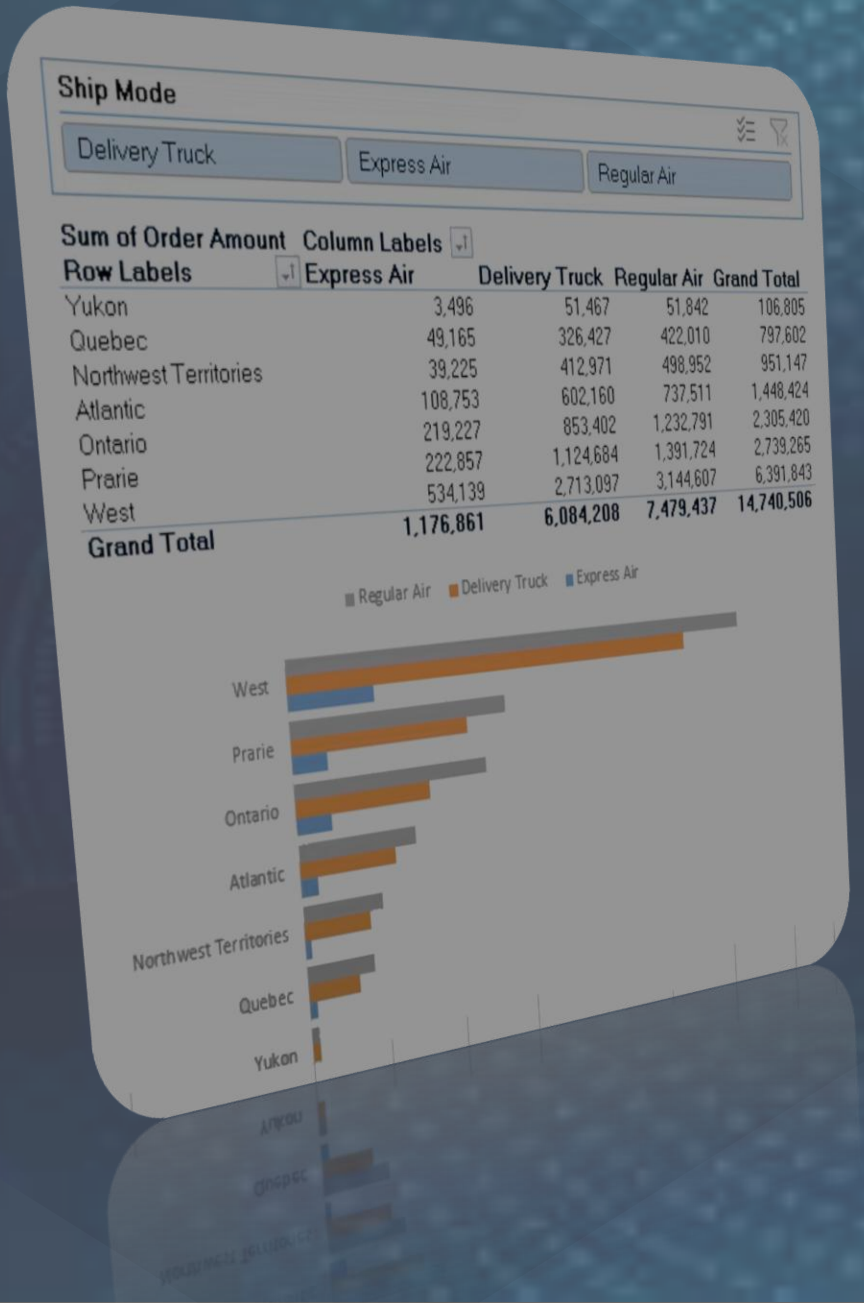
Ludovic Moreau – moreau@artofnet.ch


Ship Mode

Delivery Truck | Express Air | Regular Air

Sum of Order Amount	Column Labels			
Row Labels	Express Air	Delivery Truck	Regular Air	Grand Total
Yukon	3,496	51,467	51,842	106,805
Quebec	49,165	326,427	422,010	797,602
Northwest Territories	39,225	412,971	498,952	951,147
Atlantic	108,753	602,160	737,511	1,448,424
Ontario	219,227	853,402	1,232,791	2,305,420
Prarie	222,857	1,124,684	1,391,724	2,739,265
West	534,139	2,713,097	3,144,607	6,391,843
Grand Total	**1,176,861**	**6,084,208**	**7,479,437**	**14,740,506**

EXCEL TRAINING SESSIONS

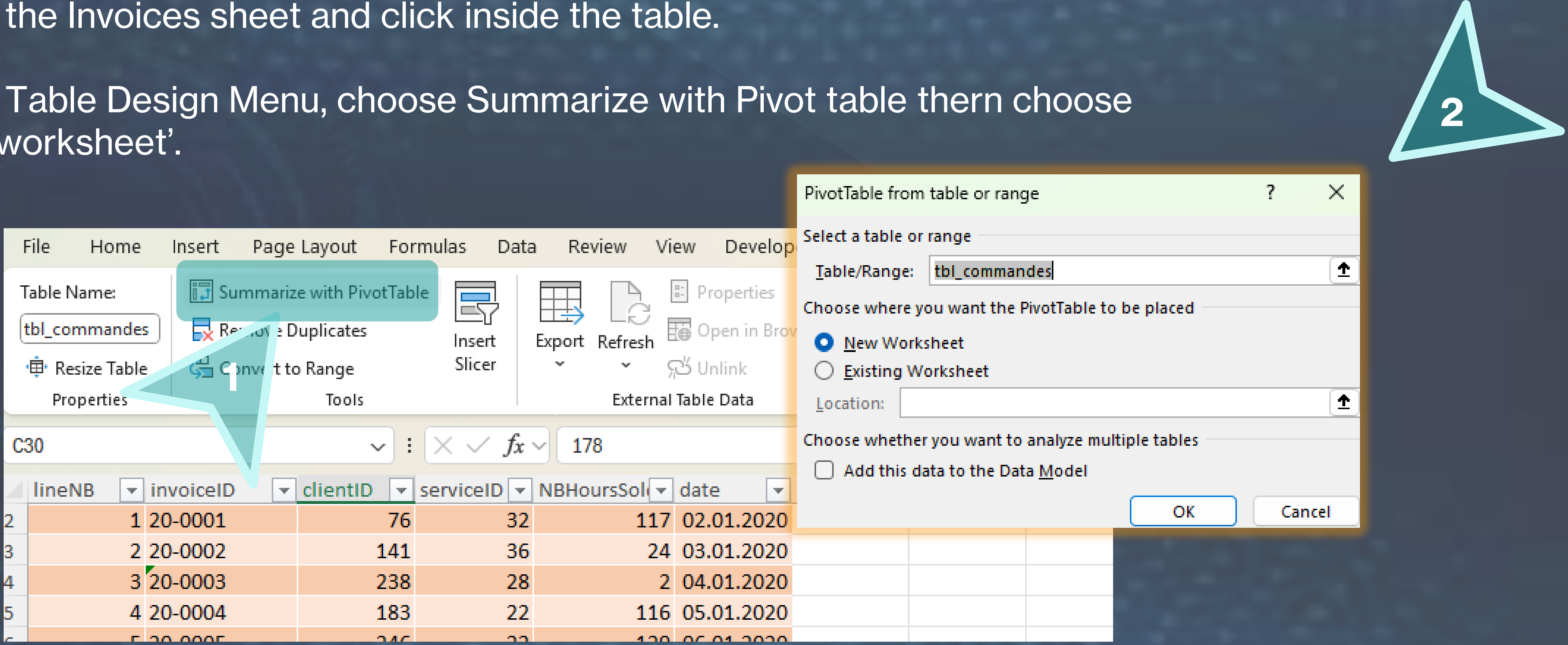
- PIVOT TABLE EXERCISE

Pivot table from a single table

Open the PP_SampleData_START.xlsx file

Go to the Invoices sheet and click inside the table.

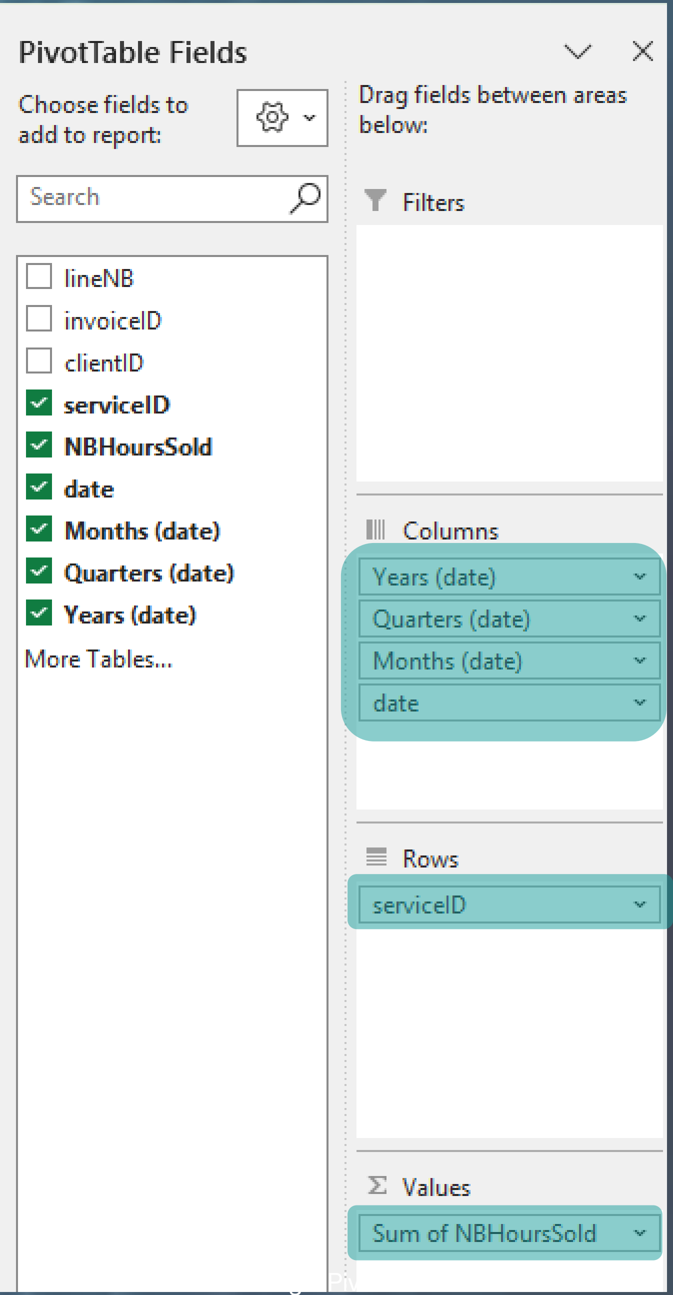
In the Table Design Menu, choose Summarize with Pivot table thern choose 'New worksheet'.



Pivot table from a single table

In the empty pivot table worksheet, use the right pane to set up the pivot table by dragging the NBHoursSold field to the values, Date to columns and serviceID to rows.

Sum of I	Columi			
	⊞2020	⊞2021	⊞2022	Grand Total
Row				
1		428	642	1070
2		628	638	1266
3		970	1232	2202
4		1050	340	1390
5		1090	1350	2440
6		1388	1554	2942
7		710	196	906
8		418	368	786
9		486	586	1072
10	612	556	440	1608
11	314	496	602	1412
12	476	500	272	1248
13	670	952	836	2458
14	1050	370	898	2318
15	698	1268	1478	3444
16	298	112	348	758
17	614	1150	1026	2790
18	994	758	928	2680



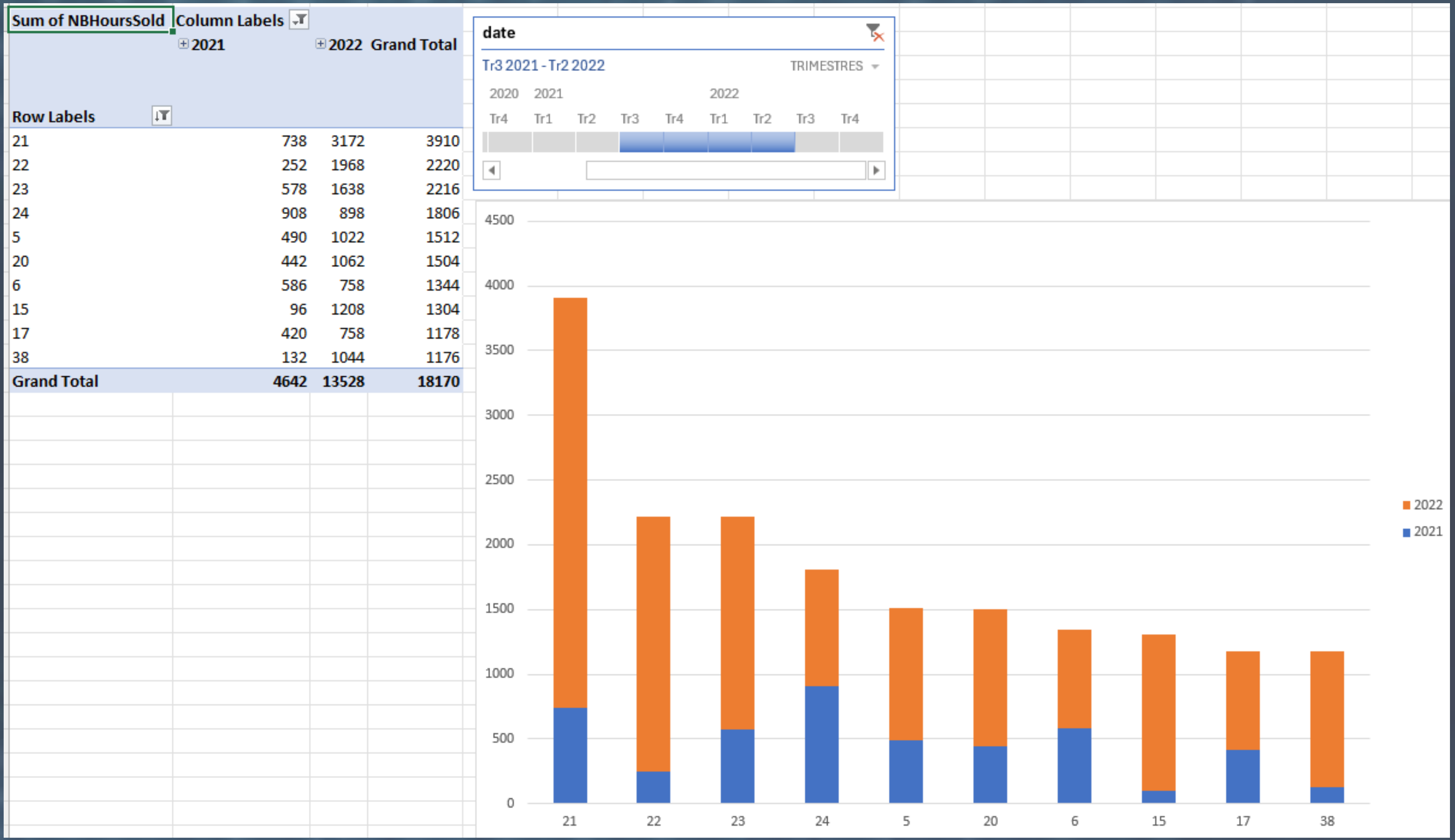
Insert a Pivot chart and a slicer

Insert a pivot chart from the pivot table (stacked columns).

Insert a timeline slicer with the date.

Sort the pivot table by the NBHoursSold measure in descending order.

Filter the top 10 values in the sorted table.



Sum of NBHoursSold	Column Labels		
	2021	2022	Grand Total
Row Labels			
21	738	3172	3910
22	252	1968	2220
23	578	1638	2216
24	908	898	1806
5	490	1022	1512
20	442	1062	1504
6	586	758	1344
15	96	1208	1304
17	420	758	1178
38	132	1044	1176
Grand Total	4642	13528	18170

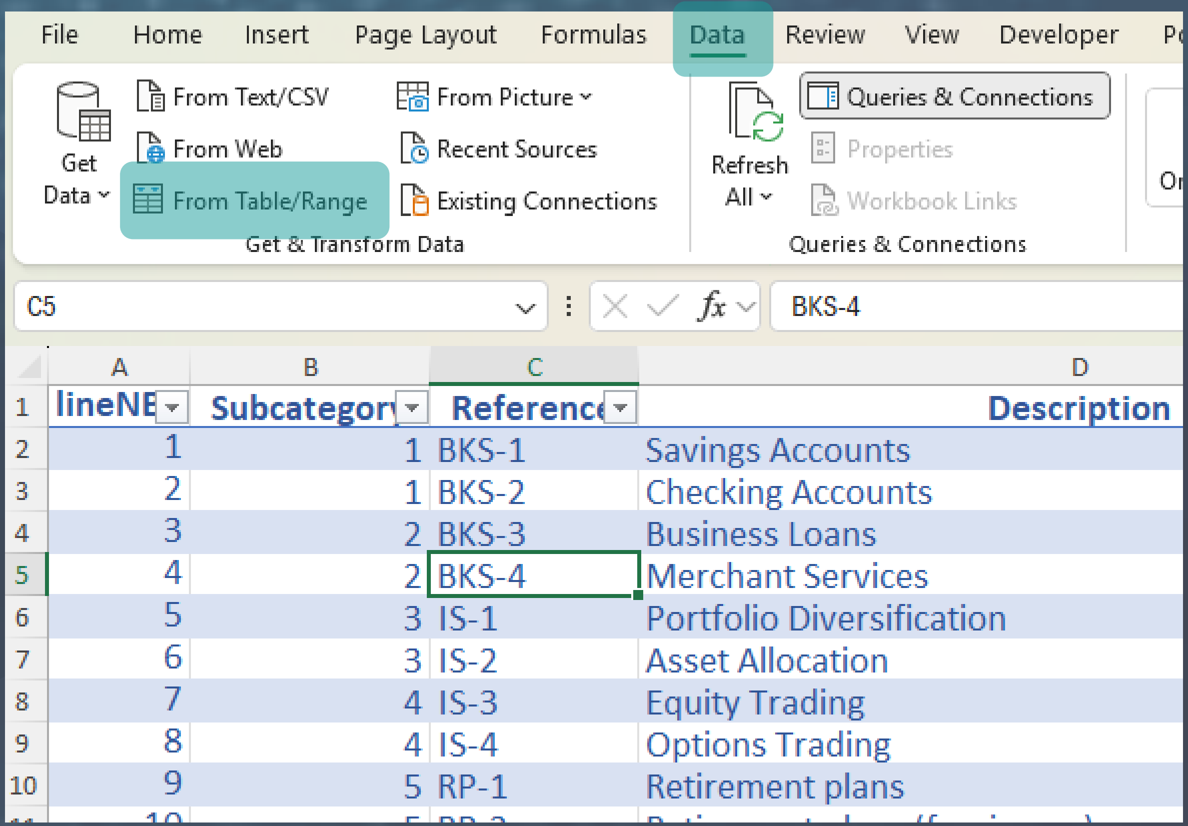
Get & prepare the Data from multiple sheets

Open the PP_SampleData_START.xlsx file

From the Services sheet, click inside the table and go to Data > From Table/Range.

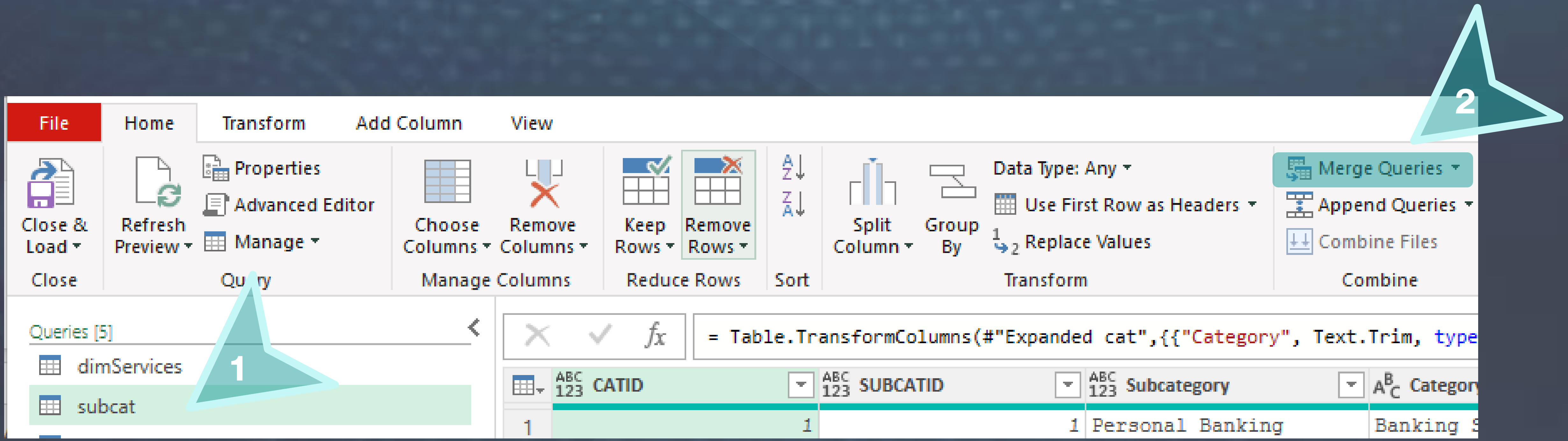
The table is loaded into the Power Query editor.

Repeat the process for the Cat and Subcat tables.



Use Power Query to merge tables

In Power Query editor, select the Subcat query and Click on Home > Merge Queries.

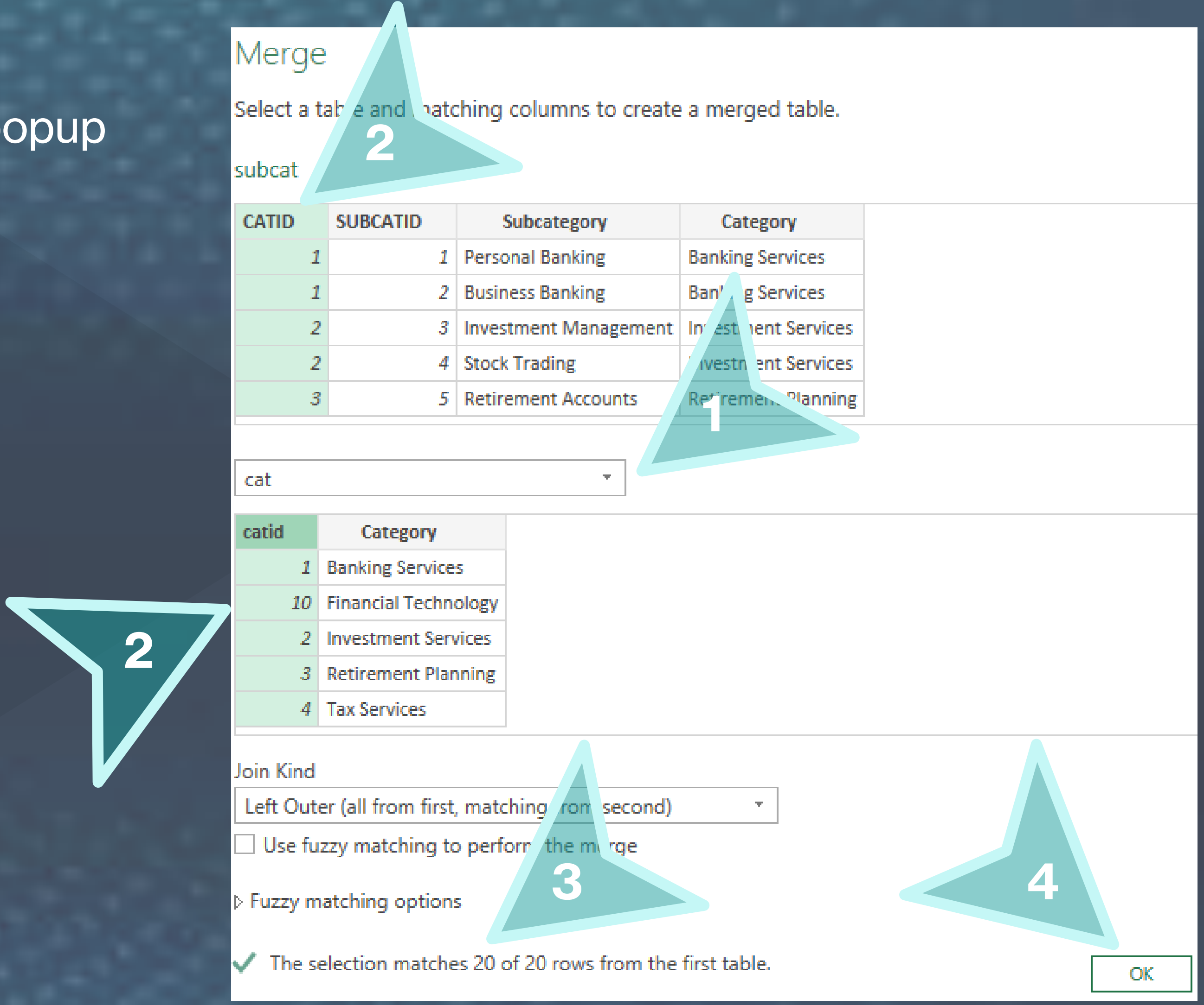


Use Power Query to merge tables

Select the Cat table as the second table in the popup window then select the common column in both tables.

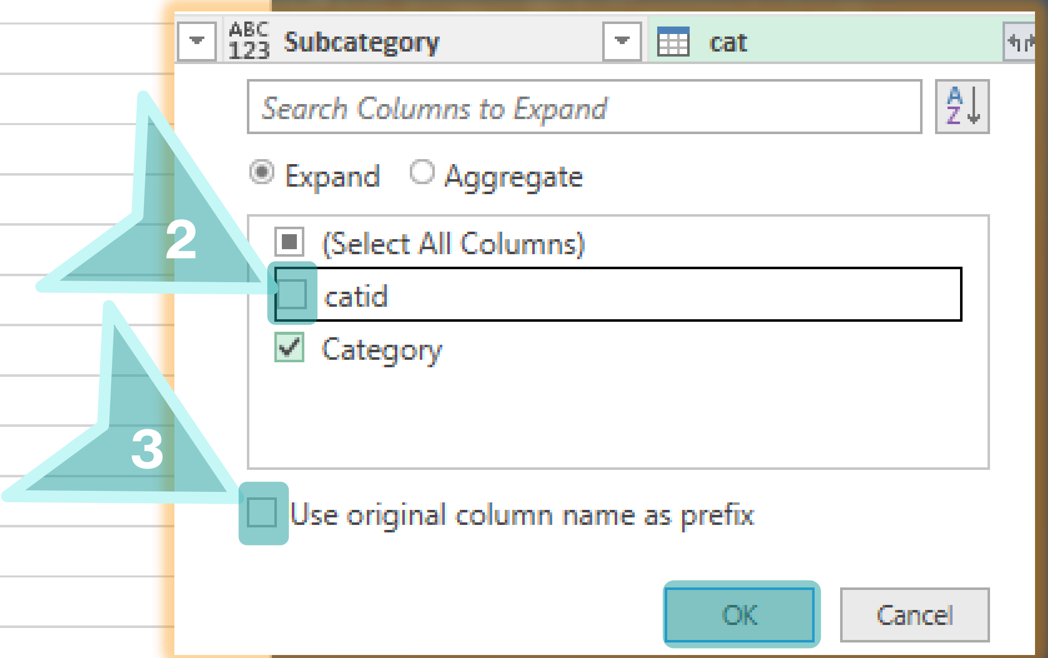
Verify that Matches are correctly made.

Click OK.

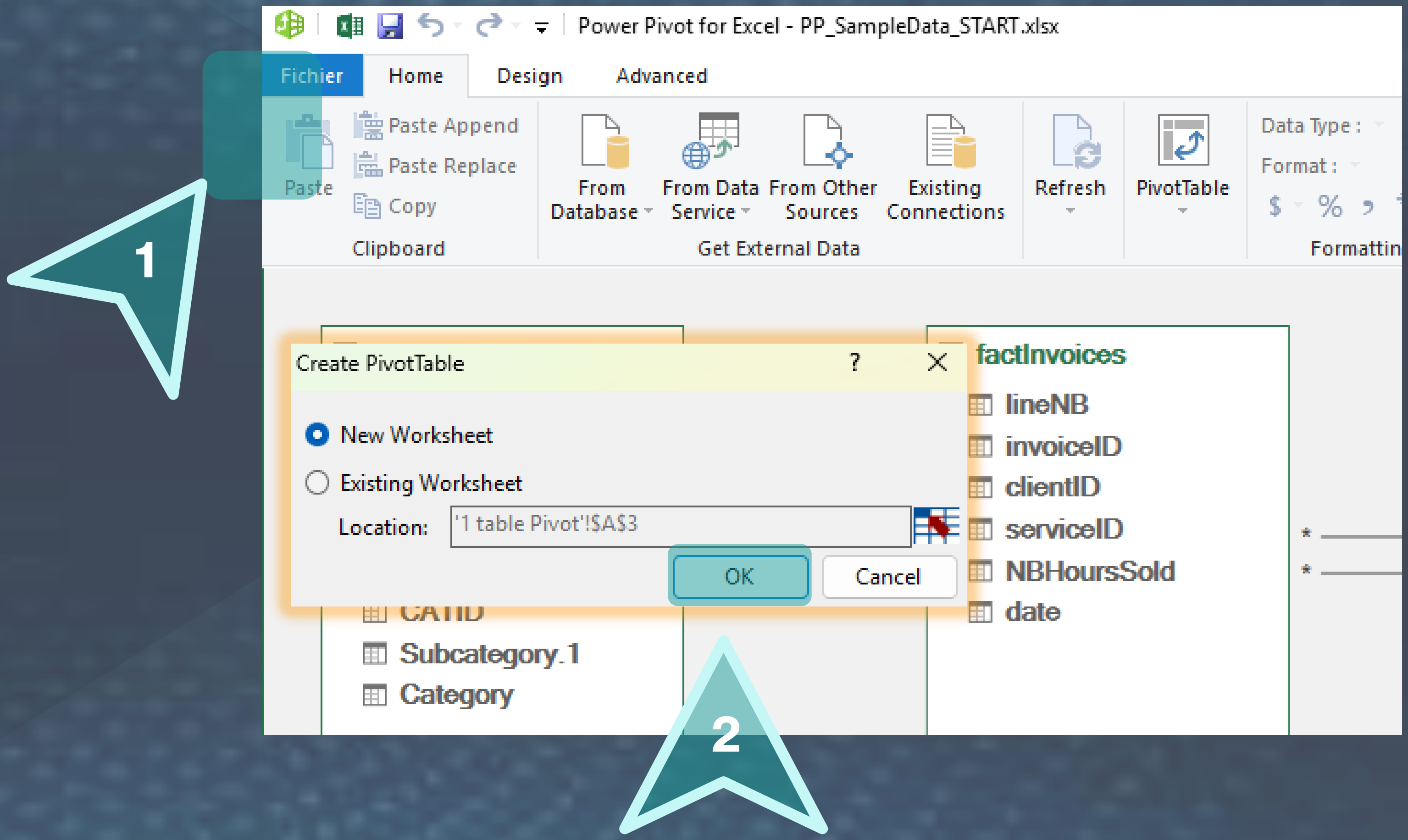


Expand merged table in query Subcat

	CATID	SUBCATID	Subcategory	cat
1	1	1	Personal Banking	Table
2	1	2	Business Banking	Table
3	2	3	Investment Management	Table
4	2	4	Stock Trading	Table
5	3	5	Retirement Accounts	Table
6	3	6	Annuities	Table
7	4	7	Individual Tax	Table
8	4	8	Business Tax	Table
9	5	9	Life Insurance	Table
10	5	10	Property & Casualty	Table
11	6	11	Wills & Trusts	Table
12	6	12	Inheritance Planning	Table
13	7	13	Financial Consulting	Table
14	7	14	Investment Advisory	Table
15	8	15	Home Loans	Table
16	8	16	Commercial Mortgages	Table
17	9	17	Credit Counseling	Table
18	9	18	Debt Management	Table
19	10	19	Online Banking	Table
20	10	20	Blockchain & Cryptocu...	Table



Import data from the CLIENTS table



Create flat file

Duplicate the cat query and change the name of the table in the source step to the name of the table in the CLIENTS sheet. Rename to dimClients.

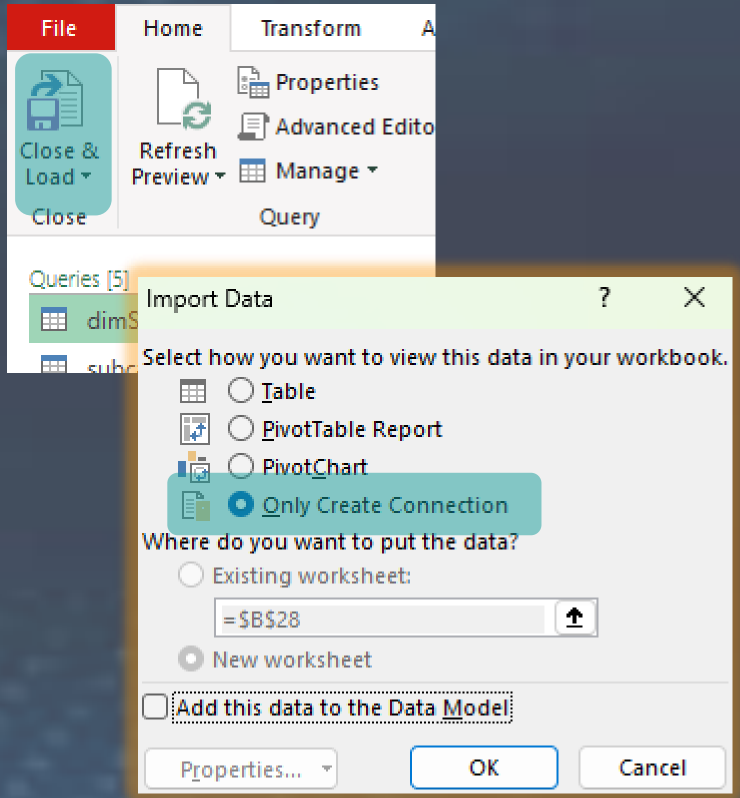
Repeat the same procedure for the INVOICES table and rename the Invoices query to factInvoices.

Merge the factInvoices and dimClients queries and add the client name, type, location and canton columns to the fact table.

Merge factInvoices and dimServices queries and add the Services Description, rate per hour, category and subcategory columns to the fact table. Rename the columns for the names os the cat and subcat.

factInvoices is now ready to be summarized into a pivot table.

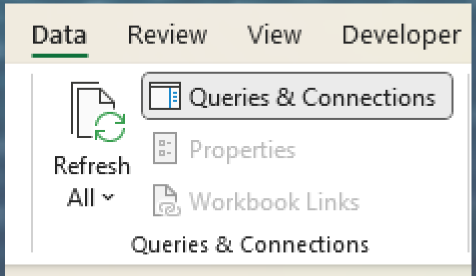
Click on Home > Close & Load > Only create connection

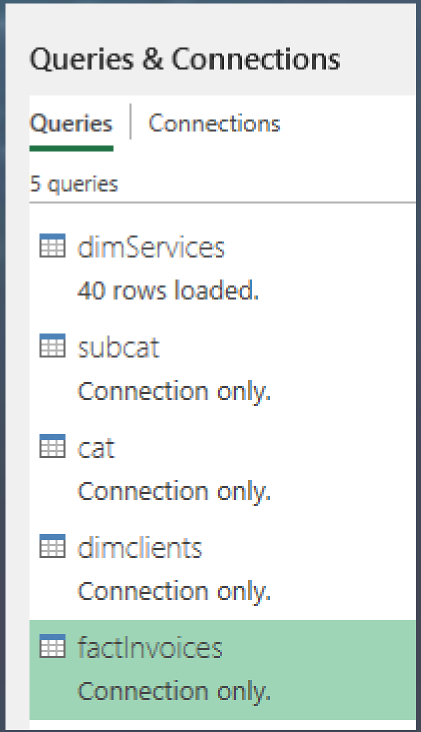


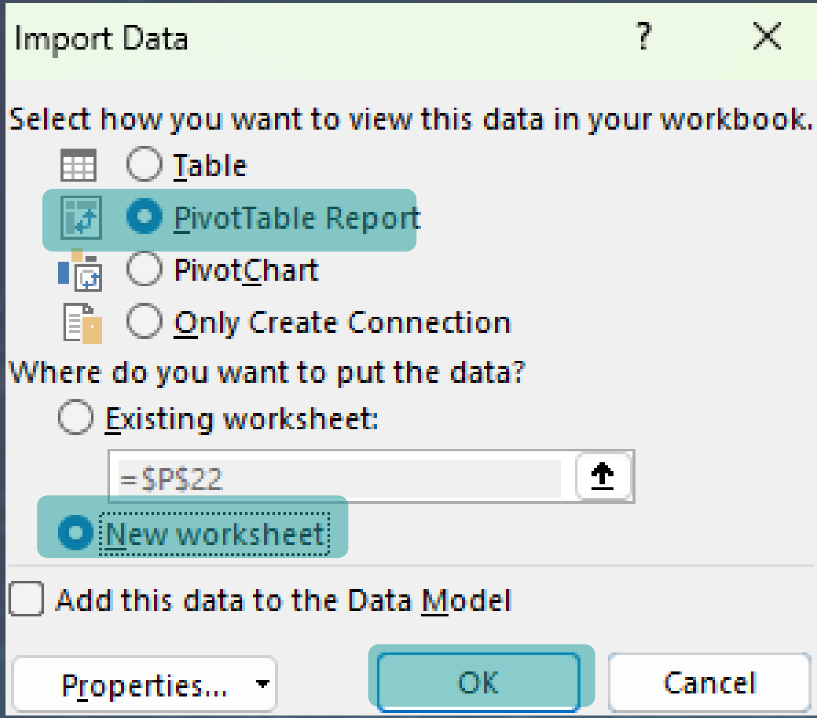
Create Pivot Table from the factInvoices query

In excel, Go to Data > Queries & Connections.

In the right pane that appears, right click on the factInvoices Query and select ‘Load to’ and choose the following options:



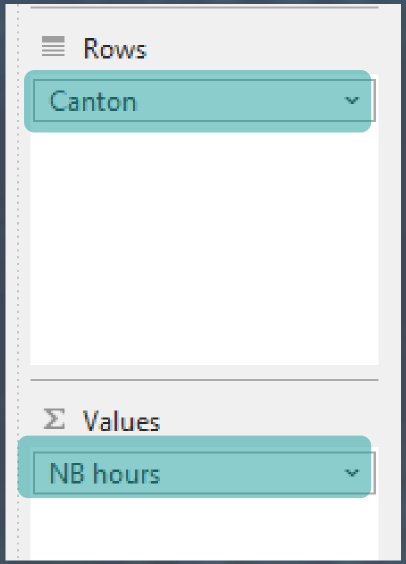




A new Pivot table is created

All the fields from the different merges are now available. Drag the NBHoursSold in the values window and the canton in the rows. You can rename the column in the pivot table if desired.

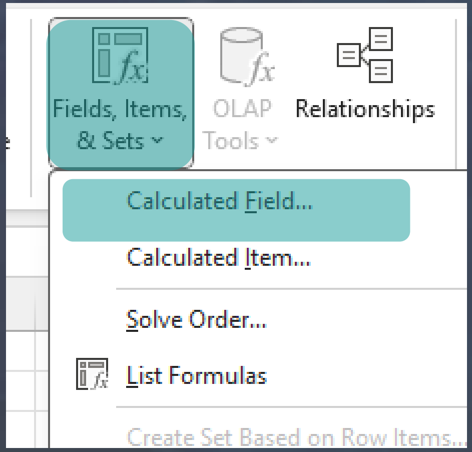
	A	B
1	Row Labels	NB hours
2	BE	2417
3	BL	438
4	BS	323
5	FR	14761
6	GE	34159
7	JU	3024
8	LU	671
9	NE	8088
10	NW	461
11	SG	52
12	TI	3924
13	VD	30342
14	VS	10742
15	ZG	1355
16	ZH	764
17	**Grand Total**	**111521**

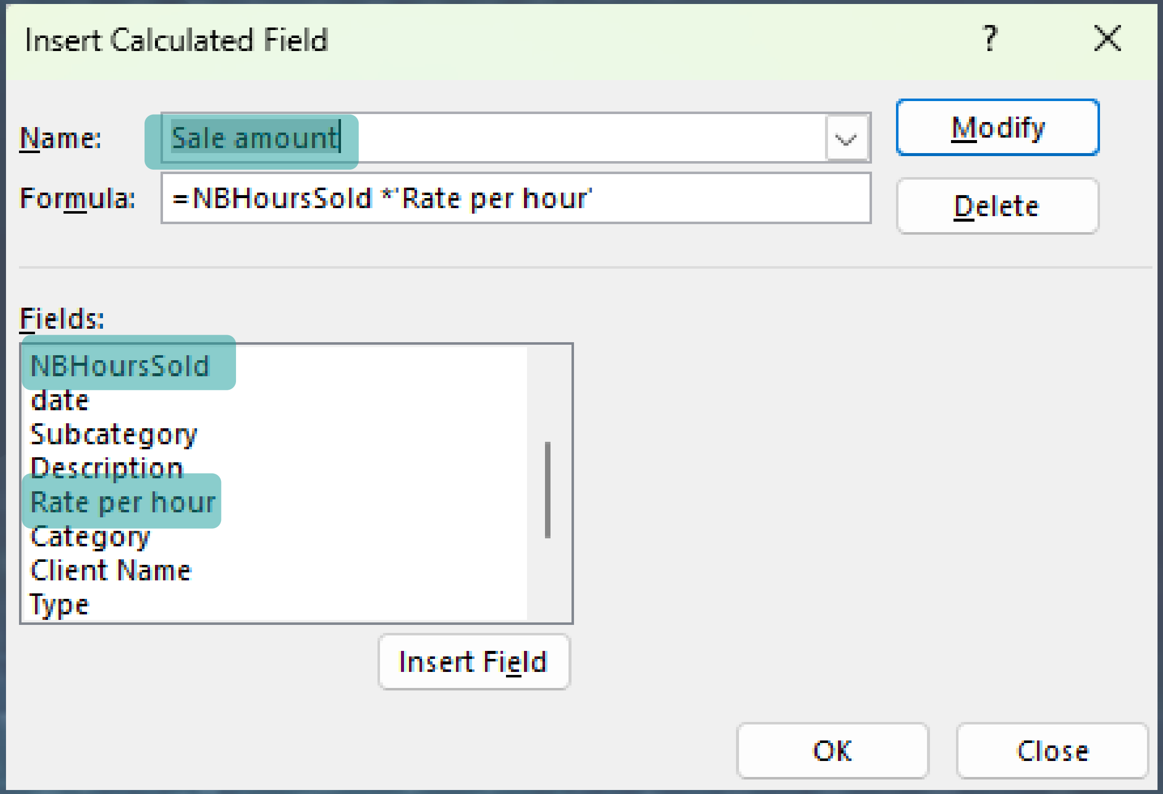


Create a calculated field

GO to PivotTable Analyze > Fields, Items & Sets > Calculated Field

Type a name for this calculated field and create the formula by clicking the fields in the fields lits separated by *.

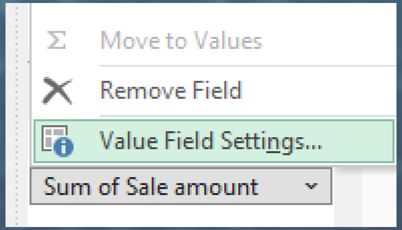




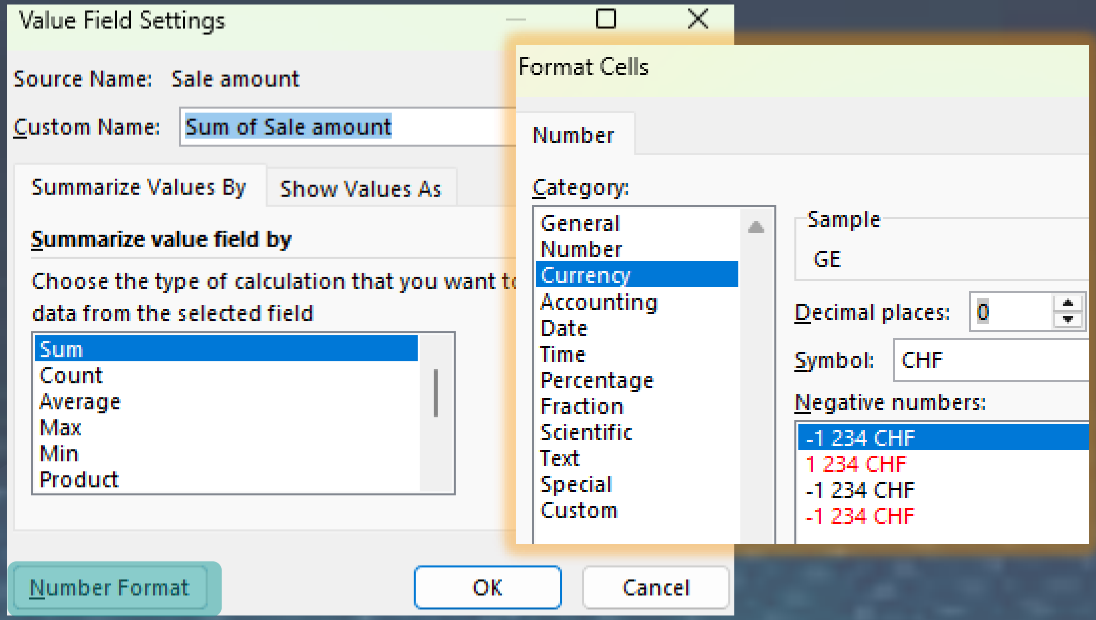
Create a calculated field

The Sale amount field is added to the pivot table.

Click on the drop down arrow on the right of the field in the Values window and choose 'Value Field Settings'.



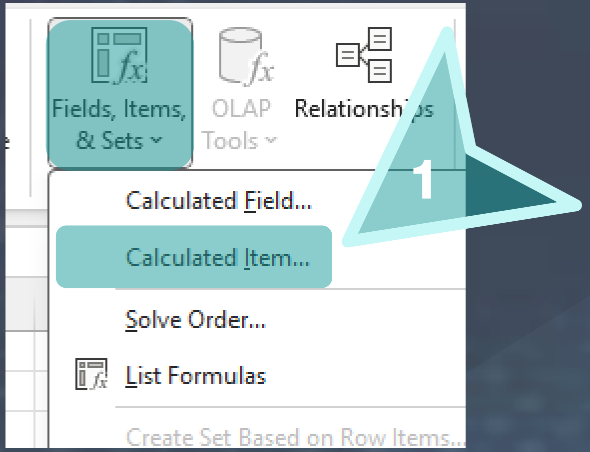
Select 'Number Format' min the popup window then format the field as currency with 0 decimal.

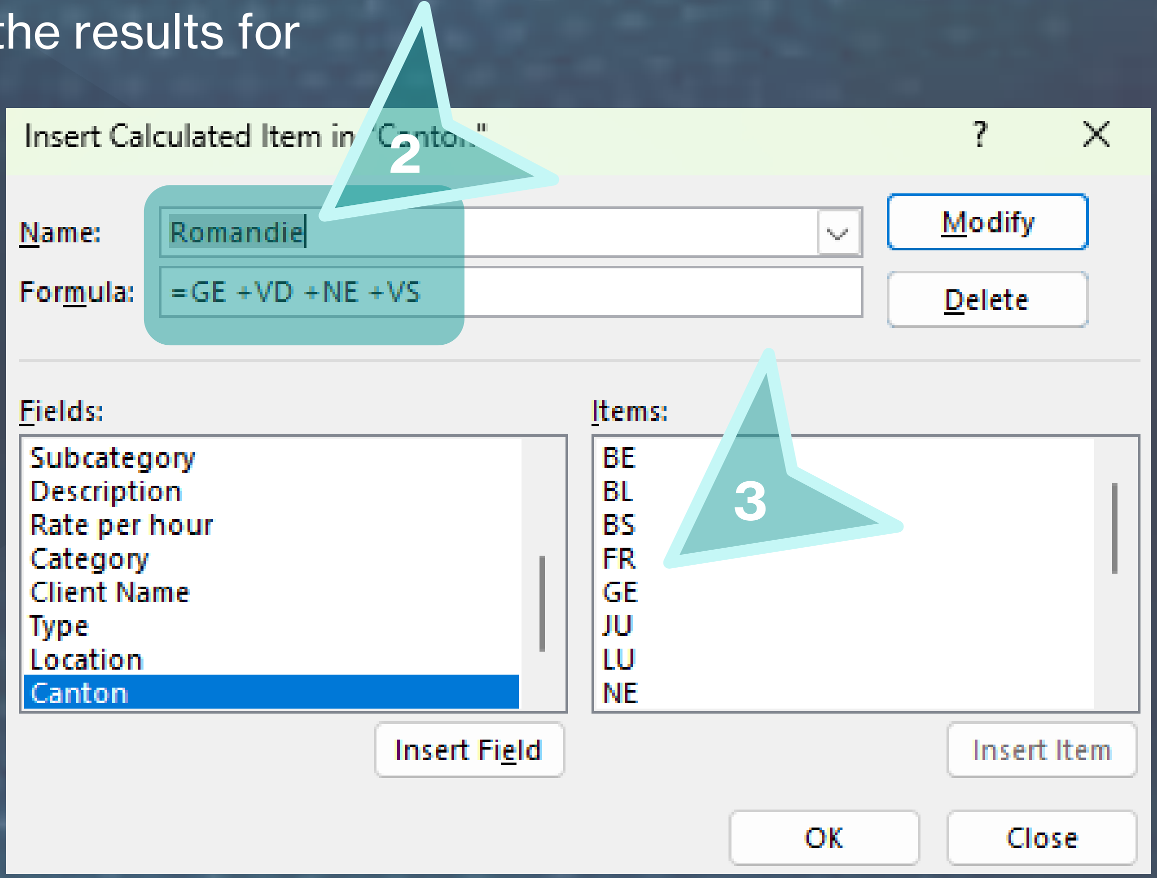


Create a calculated item

GO to PivotTable Analyze > Fields, Items & Sets > Calculated Item.

Type a name then create the formula that adds the results for several cantons.





Manual sort of the table

You can drag fields in the 1st column to reorder them within the pivot table

For example, drag ‘Romandie’ and ‘VD’ in the top 2 rows of the pivot table.

	A	B	C
1	Row Labels	NB hours	Sum of Sale amount
2	Romandie	83331	11 564 676 180 CHF
3	VD	30342	1 586 583 180 CHF
4	BE	2417	9 160 430 CHF
5	BL	438	302 220 CHF
6	BS	323	177 650 CHF
7	FR	14761	350 868 970 CHF
8	GE	34159	1 943 305 510 CHF
9	JU	3024	12 216 960 CHF
10	LU	671	671 000 CHF
11	NE	8088	108 945 360 CHF
12	NW	461	322 700 CHF
13	SG	52	7 280 CHF
14	TI	3924	26 997 120 CHF
15	VS	10742	173 268 460 CHF
16	ZG	1355	2 533 850 CHF
17	ZH	764	1 230 040 CHF

Design of the pivot table

Copy the entire sheet.

In the rows window, replace the cantons by the Category and add the subcategory and the description to create an automatic hierarchy.

Use the Design tab of the pivot table to display / hide totals and subtotals. Add blank rows and choose the layout type.

Select your favorite color style.

Rename the pivot table and the sheet to Pvt_CatSubCatDesc. If not done yet, repeat for all current and future pivot tables.

	A	B	C
1	Row Labels	NB hours	Sum of Sale amount
2	⊟ Banking Services	5928	55 486 080 CHF
3	⊟ Business Banking	3592	20 689 920 CHF
4	Business Loans	2202	8 455 680 CHF
5	Merchant Services	1390	2 668 800 CHF
6			
7	⊟ Personal Banking	2336	8 409 600 CHF
8	Checking Accounts	1266	2 532 000 CHF
9	Savings Accounts	1070	1 712 000 CHF
10			
11	⊟ Credit & Debt	9732	126 516 000 CHF
12	⊟ Credit Counseling	4650	32 550 000 CHF
13	Credit Score Improvement	1088	1 523 200 CHF
14	Debt Settlement	3562	19 947 200 CHF
15			
16	⊟ Debt Management	5082	30 492 000 CHF
17	Credit Repair Services	2380	7 140 000 CHF
18	Debt Consolidation Plans	2702	8 106 000 CHF
19			
20	⊟ Estate Planning	25820	1 009 562 000 CHF
21	⊟ Inheritance Planning	11634	193 124 400 CHF
22	Beneficiary Designation	5538	38 766 000 CHF
23	Estate Tax Planning	6096	58 521 600 CHF
24			
25	⊟ Wills & Trusts	14186	319 185 000 CHF

Create a date hierarchy

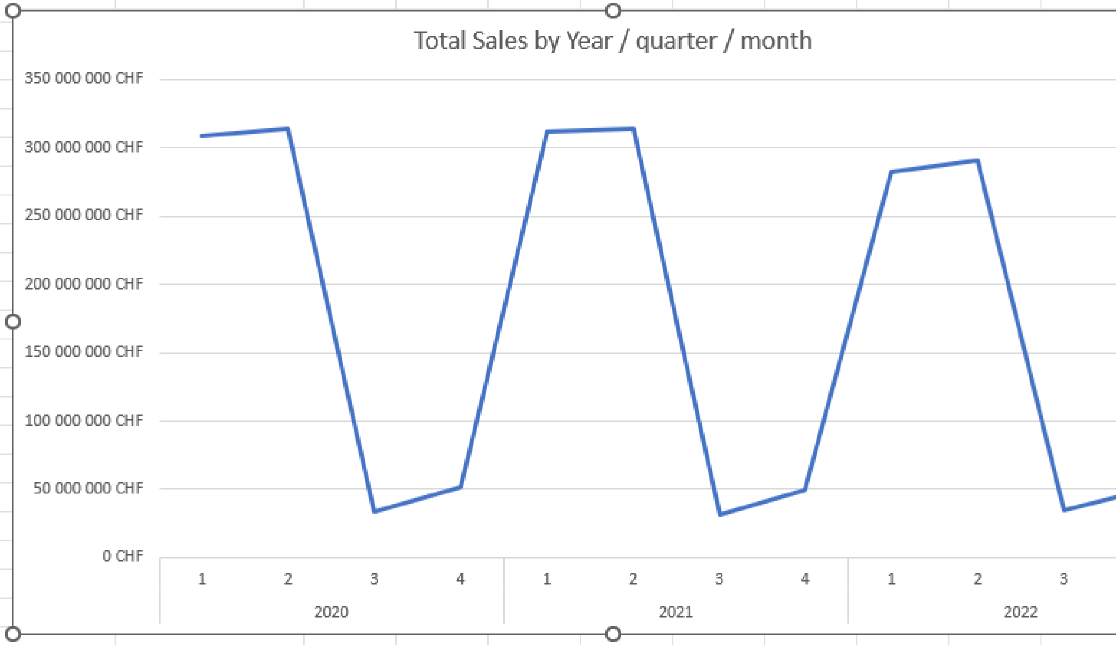
Copy the entire sheet.

Go to Power Query and in the factInvoices query, create the Year, quarter, Month and Month Name columns by adding date specific columns.

In the copied sheet, drag the Year, quarter and Month Name in the Rows. Keep only Sale amount in the values.

Insert a pivot chart (line chart) that displays to sales with the granularity given by the level in the date hierarchy.

Row Labels	Sum of Sale amount
⊟ 2020	2 328 871 820 CHF
⊞ 1	308 617 800 CHF
⊞ 2	314 003 280 CHF
⊞ 3	33 642 720 CHF
⊞ 4	51 305 760 CHF
⊟ 2021	2 306 804 400 CHF
⊞ 1	311 535 000 CHF
⊞ 2	314 292 160 CHF
⊞ 3	31 883 360 CHF
⊞ 4	48 862 440 CHF
⊟ 2022	2 195 702 640 CHF
⊞ 1	282 467 000 CHF
⊞ 2	290 814 160 CHF
⊞ 3	35 001 920 CHF
⊞ 4	49 922 680 CHF



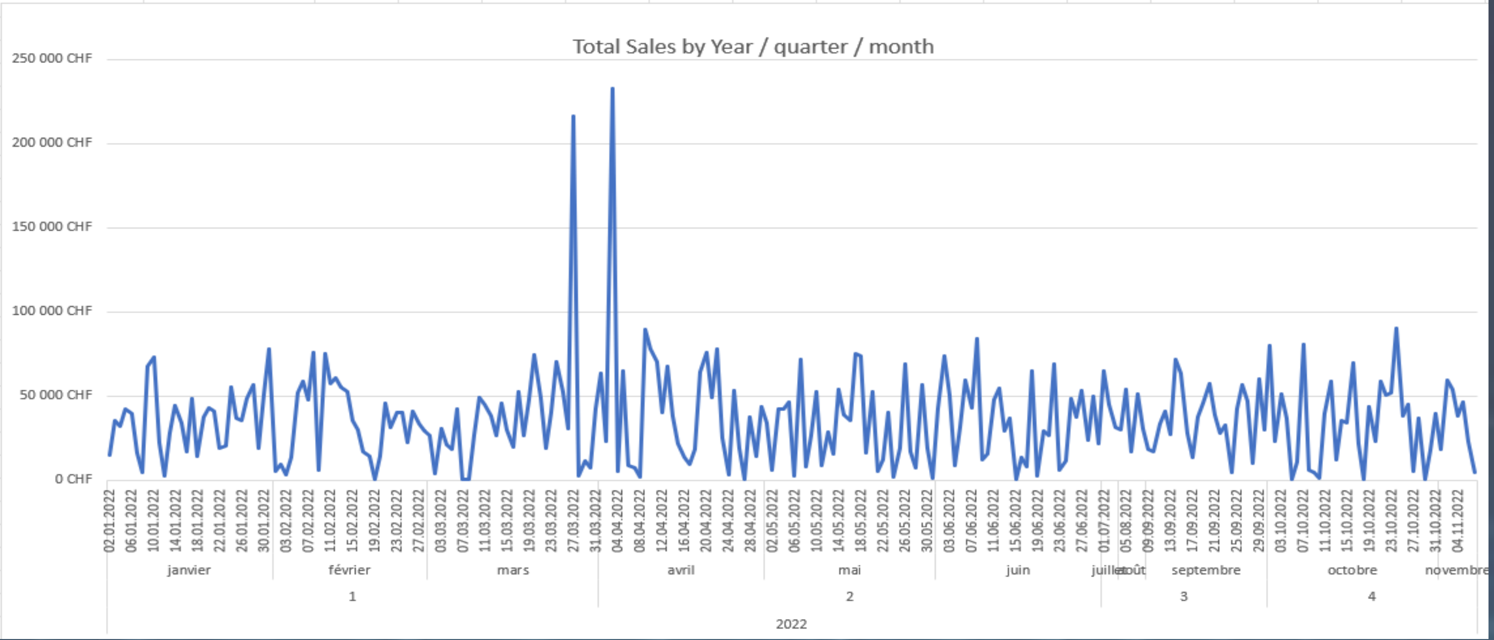
Analyze daily sales

Add the date field to the rows

Expand the date hierarchy to the date level.

Insert a slicer on Year and click on the desired year to view the data only for that specific year.

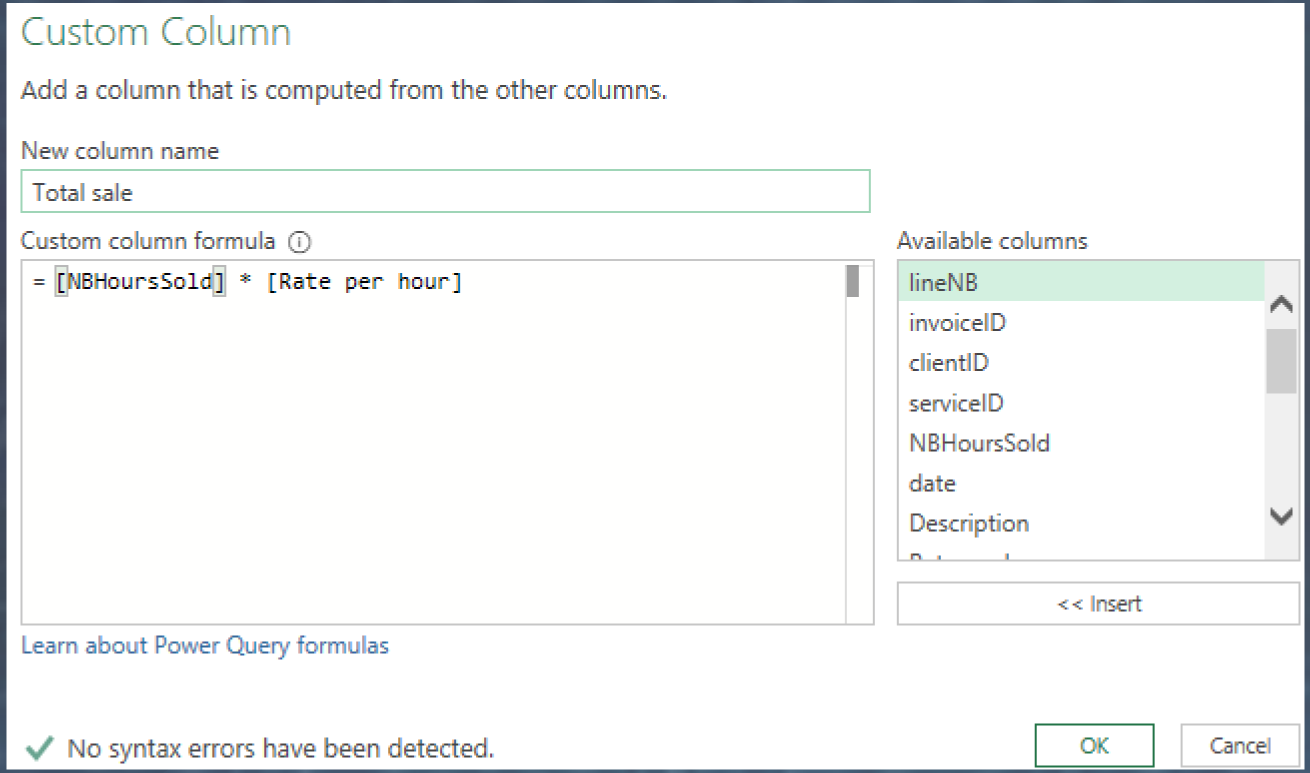
Row Labels	Sum of Sale amount
⊟ 2022	2 195 702 640 CHF
⊟ 1	282 467 000 CHF
⊟ janvier	32 837 200 CHF
02.01.2022	14 880 CHF
03.01.2022	35 280 CHF
04.01.2022	31 920 CHF
05.01.2022	42 400 CHF
06.01.2022	39 600 CHF
07.01.2022	16 400 CHF
08.01.2022	4 320 CHF
09.01.2022	67 800 CHF
10.01.2022	73 200 CHF
11.01.2022	22 400 CHF
12.01.2022	2 400 CHF
13.01.2022	27 600 CHF
14.01.2022	44 240 CHF
15.01.2022	34 080 CHF
16.01.2022	16 800 CHF
17.01.2022	48 400 CHF
18.01.2022	14 000 CHF
19.01.2022	37 200 CHF
20.01.2022	43 200 CHF
21.01.2022	40 800 CHF
22.01.2022	19 200 CHF
23.01.2022	20 160 CHF
24.01.2022	55 600 CHF
25.01.2022	36 800 CHF
26.01.2022	35 600 CHF
27.01.2022	48 400 CHF
28.01.2022	56 400 CHF
29.01.2022	18 720 CHF
30.01.2022	50 800 CHF



Show values as % and add cond. formatting

Copy the last sheet and drag the Category name in the rows and the Total Sales in the values.

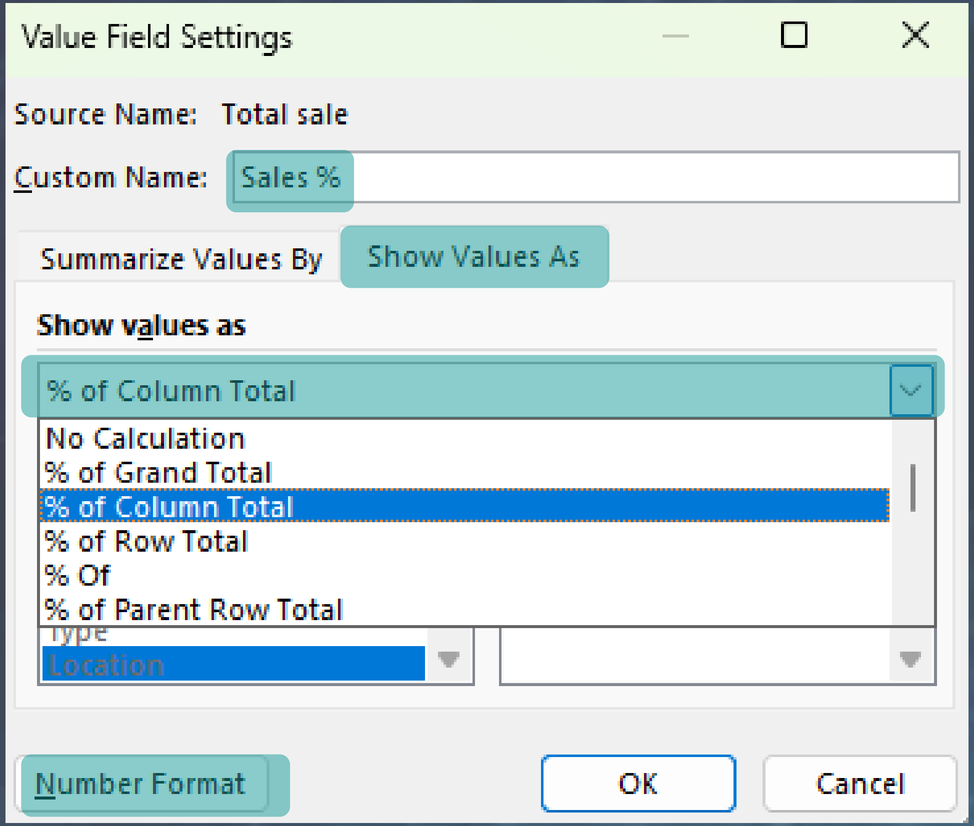
Create the Total Sales field in Power Query by adding a custom column with the multiplication of the NBHoursSold by The Rate per Hour.



Show values as % and add cond. formatting

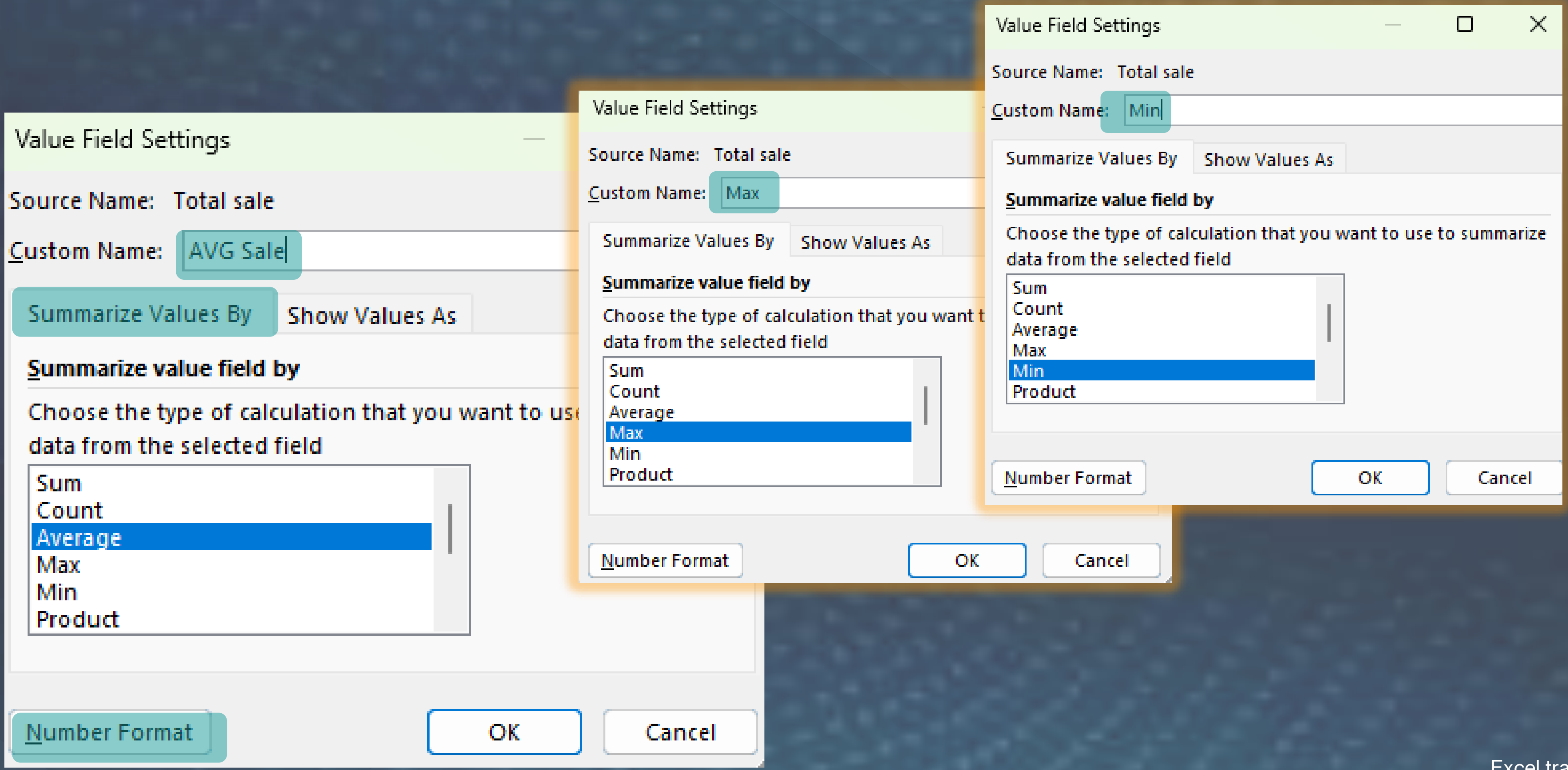
Insert the Total Sales field 5 times in the values window.

For the second, go to Field Value Settings, change the name and Choose 'Show Values As' and select '% of Column Total'. Format to percentage.



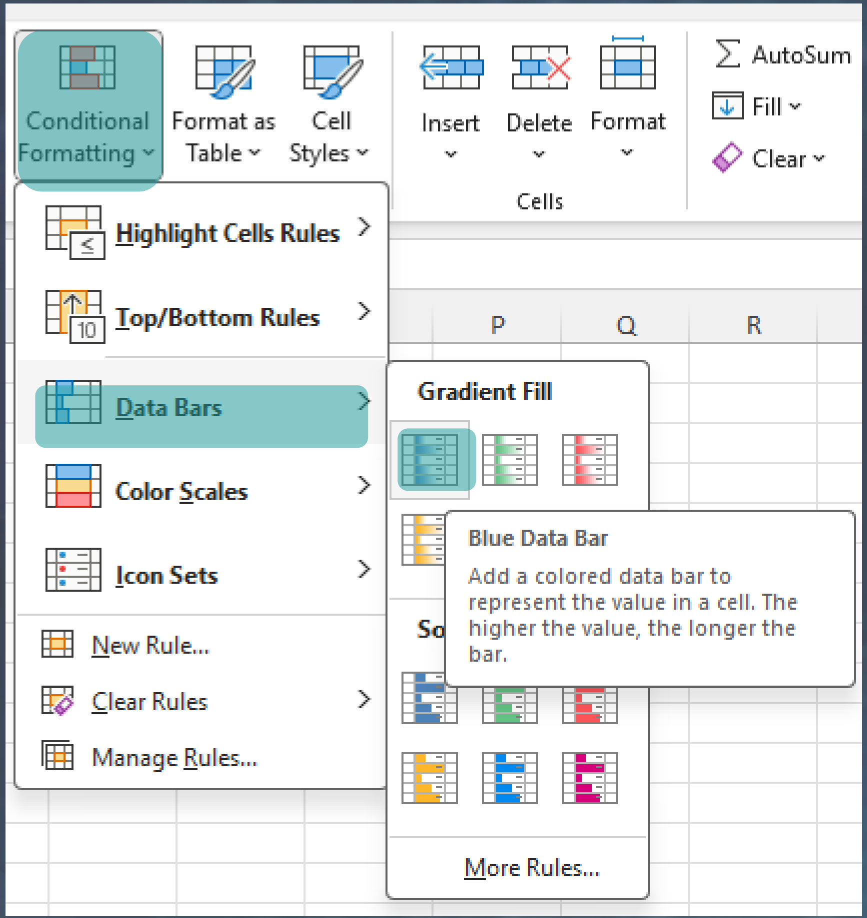
Show values as % and add cond. formatting

For the last 3 values of Total, change the summarization to average, maximum and minimum respectively.

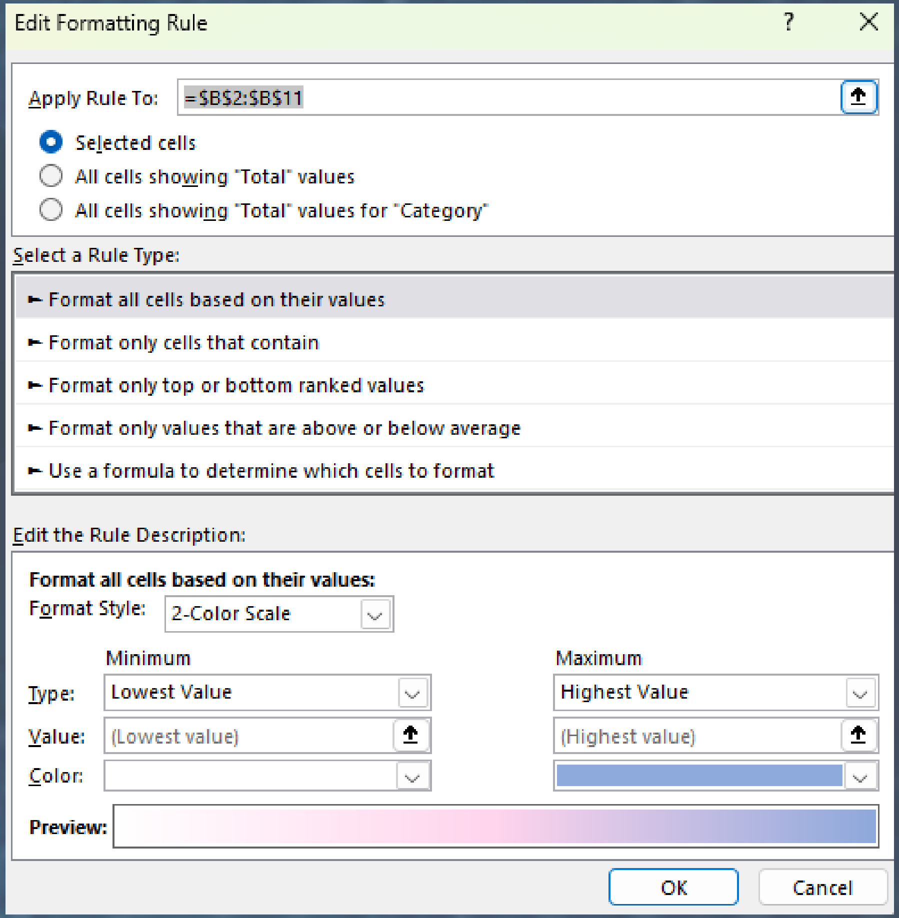


Show values as % and add cond. formatting

Apply conditional formatting to the Sales % column with a data bar.



Apply conditional formatting to the Total Sales column with a color scale and personalize the colors.



Final pivot table

The final Pivot Table should look like this. Insert a slicer on Year, Client type and Cantons to filter the results as you wish.

Row Labels	Total	Sales %	AVG sale	Max	Min
Banking Services	664 640 CHF	4.82%	7 912 CHF	17760	360
Credit & Debt	973 200 CHF	7.06%	7 486 CHF	15000	400
Estate Planning	2 973 220 CHF	21.58%	8 797 CHF	21000	150
Financial Advisory	2 085 540 CHF	15.14%	12 414 CHF	27000	360
Financial Technology	1 321 360 CHF	9.59%	13 764 CHF	29200	420
Insurance Services	1 346 000 CHF	9.77%	8 108 CHF	17520	200
Investment Services	775 080 CHF	5.63%	8 074 CHF	19600	100
Mortgage Services	1 696 580 CHF	12.32%	7 747 CHF	17280	120
Retirement Planning	640 800 CHF	4.65%	9 154 CHF	17520	120
Tax Services	1 298 940 CHF	9.43%	10 647 CHF	21600	560